

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Assignment on **Genetic Algorithms**:

Q1. Solution:

Generation 1 - Reproduction Process:

i	Initial Population	Value x_i	Fitness $f(x_i)$	$\frac{f(x_i)}{\sum_{i=1}^6 f(x_i)}$	Roulette Wheel Copies	New Parents
1	001.0000	1.00	0.1600	0.0077	1	1.00
2	010.0000	2.00	0.4442	0.0215	1	2.00
3	011.0100	3.25	15.7480	0.7609	2	3.25
4	100.0000	4.00	3.9841	0.1925	1	4.00
5	101.1000	5.50	0.2499	0.0121	1	5.50
6	110.1000	6.50	0.1111	0.0054	0	3.25

Fitness Sum = **20.6974**; Fitness Average = **3.4496**